

CSA1304-THEORY OF COMPUTATION LAB EXPERIMENTS

6. Write a C program to check whether a given string belongs to the language defined by a Context Free Grammar (CFG)

$$S \rightarrow 0S1 \mid \epsilon$$

AIM

Write a C program to check whether a given string belongs to the Context-Free Grammar (CFG).

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char str[100];
    int i, len;
    int zero = 0, one = 0;

    printf("Enter the binary string: ");
    scanf("%s", str);

    len = strlen(str);

    i = 0;

    while(i < len && str[i] == '0')
    {
        zero++;
        i++;
    }

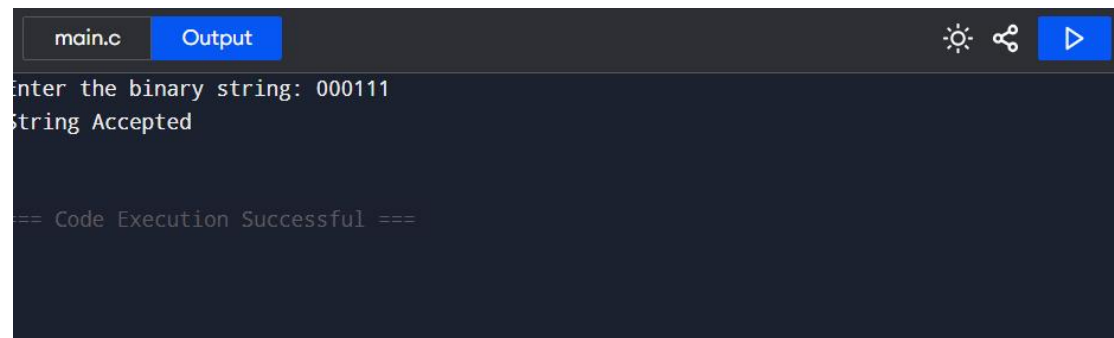
    while(i < len && str[i] == '1')
    {
        one++;
        i++;
    }

    if(i == len && zero == one)
        printf("String Accepted\n");
}
```

```
    else
        printf("String Rejected\n");

    return 0;
}
```

OUTPUT



```
main.c  Output
Enter the binary string: 000111
String Accepted

=== Code Execution Successful ===
```